Install the Training Academy

This package installs the website, backend, database, browser terminals, eleven classes with quizzes and exams, two Academy assessments, six games and four labs. Users and results start fresh. The video library is excluded; the built-in lab tutorial is included.

For learner PCs running Ubuntu, use `INSTALL-TROUBLESHOOTING-UBUNTU.pdf` for the updated DNS and certificate steps. The existing `INSTALL-TROUBLESHOOTING.pdf` is retained as the previous version.

Install from a USB drive

1. Prepare the drive. Use a 32 GB or larger exFAT USB drive labelled ACADEMY by default. If your drive has another label, keep it; after connecting, check `lsblk -f` and substitute its exact `LABEL` in step 3. Formatting erases a drive, so save its existing files first. Copy the entire GDP Installer folder onto it. FAT32 cannot hold the large files. Copying only the launcher will not work.
2. Plug it into the Proxmox server. Proxmox VE 9 must already be installed. Use a dedicated, empty, standalone node with no VMs or containers. Open Proxmox web interface → your node → Shell and sign in as root.
3. Paste this whole block into the shell:

   ```
   mkdir -p /mnt/academy
   mount -t exfat -o ro /dev/disk/by-label/ACADEMY /mnt/academy
   bash '/mnt/academy/GDP Installer/INSTALL.sh'
   ```

   If the drive is already mounted there, run the last line. If it has another label, check `lsblk -f` and replace only `ACADEMY` in the mount path with the exact `LABEL`. For any other mount error, check the drive label and connection before continuing. Proxmox has no desktop file manager, so its direct-server method uses this command instead of double-clicking a file.
4. Enter the website name and confirm. For example: `academy.training.internal`. Do not enter an IP address, `https://`, or a name ending in `.local`. The installer reads Proxmox's network and storage and checks available resources. It keeps the Academy's VM sizes and does not ask you to allocate RAM or size the lab VMs. It asks about a setting only if it cannot choose safely. Review the summary, then type yes.
5. Leave the USB connected until completion. Everything from this point installs automatically. The installation continues if you close the progress window. To check it again, run the last command from step 3. A successful finish displays Academy installation complete and a receipt. Then complete the first-sign-in steps below.

First sign-in

1. Add one LAN DNS A record. Map the website hostname to the Proxmox IP shown in the completion receipt on the DNS server/router used by the Ubuntu learner PCs. The installer cannot change that device. No public DNS or Cloudflare tunnel is needed. Learner PCs must be on the permitted LAN subnet in the installer summary. On an Ubuntu learner PC, verify the name resolves to the receipt IP:

```
getent ahosts academy.training.internal
```

   Replace the example with the hostname in your receipt. If it returns no address or the wrong address, correct the LAN DNS A record before proceeding.

2. Trust the Academy certificate on each Ubuntu learner PC. On the Proxmox root shell, record the SSH fingerprint and CA file checksum:

```
ssh-keygen -lf /etc/ssh/ssh_host_ed25519_key.pub
sha256sum /root/academy-install/academy-ca.crt
```

   On each Ubuntu PC, copy the public CA certificate. Replace `PROXMOX_LAN_IP` with the receipt IP:

```
scp root@PROXMOX_LAN_IP:/root/academy-install/academy-ca.crt .
```

   At the first SSH prompt, compare the offered SHA256 fingerprint with the Proxmox fingerprint above. Stop if it differs. Then compare the local checksum with the Proxmox checksum:

```
sha256sum academy-ca.crt
```

   Install the verified public CA into Ubuntu's system trust store:

```
sudo install -o root -g root -m 0644 academy-ca.crt /usr/local/share/ca-certificates/academy-ca.crt
sudo update-ca-certificates
```

   Restart the browser and open the exact hostname from the receipt. If Firefox still reports an untrusted issuer, open Settings → Privacy & Security → Certificates → View Certificates → Authorities → Import, select `academy-ca.crt`, and enable Trust this CA to identify websites. Restart Firefox. Never bypass a certificate warning. Copy only the `.crt` file, never a `.key` file.

3. Read the initial login in the Proxmox root shell:

```
qm guest exec 9000 -- /bin/cat /opt/training-academy/secrets/initial-admin.txt
```

   The username and password appear in the output's `out-data` field. Sign in using the receipt hostname and change the administrator password in Administration. Create learner accounts there. Keep the login details private.

4. Check the new site before admitting learners. Open classes and games. Launch and end each lab; confirm both Callback panes show different hosts and accept input. Check exercise feedback and grading, and verify two learners receive separate environments. You can adjust lab VM settings in Administration.
5. Remove the USB safely after installation has completed:

```
umount /mnt/academy
```

The LAN DNS record and Ubuntu client certificate trust are the two external setup tasks. The package automatically installs and configures the Academy itself.

Optional: double-click from Windows instead

Leave the USB in a Windows PC on the new server's LAN and double-click INSTALL-WINDOWS.cmd. Windows OpenSSH Client must be installed. Enter the Proxmox IPv4 address, compare the SSH fingerprint as above, and enter the root password only when OpenSSH asks. Up to three password prompts are normal; the password is not saved.

The launcher verifies and copies the complete package, then starts the same installer. Keep the PC awake through copying and confirmation. After confirming, installation runs on the server even if the progress connection closes. Complete the same DNS, certificate and first-login steps above. The protected copy remains under `/root/gdp-install` on Proxmox.

If setup stops

In the Proxmox root shell, check:

```
cat /root/academy-install/state /root/academy-install/step
```

`complete` means installation finished. `starting` or `running` means it is still working. `failed` means it stopped. The full log is `/root/academy-install/install.log`, and the completion receipt is `/root/academy-install/receipt.txt`.

Running INSTALL.sh again shows the saved status; it never retries a partial installation. Preserve the log, `/root/academy-media`, `/root/academy-bootstrap`, and the created VMs. Do not delete VMs or networking to force another run. Obtain help with the exact error before changing the failed installation; share logs only privately because they can contain deployment details.

If the Windows copy failed before installation started, preserve the incomplete `/root/gdp-install` directory under another unused name before copying again. Do this only if `/root/academy-bootstrap` and `/root/academy-install` do not exist and no installation service is running. The preserved copy still uses disk space.

Hardware and defaults

The installer checks suitability automatically. Plan for at least 32 GB RAM, 250 GB free LVM-thin or ZFS VM storage, and 40 GiB free under `/root`. The Windows method additionally needs enough space for the whole package. ZFS ARC reservations can require more physical RAM. A minimum-size server normally supports about two simultaneous labs; the existing ten-lab default is reduced when measured resources cannot safely support it. Two labs per learner remain the default.

VM defaults match the Academy: services 4 vCPUs / 8 GiB / 50 GiB; Netlab and Linux 2 vCPUs / 4 GiB each; Callback 2 vCPUs / 4 GiB per VM; Docker 2 vCPUs / 8 GiB. Lab disks come from the included templates. This package does not resize the Proxmox host. SSD storage improves installation and lab startup speed.